Examiner only

6. (a) Discuss the energy changes involved in a bungee jump over one complete jump (i.e. from the initial jump height to the bottom of the jump and back up again). [6 QER]

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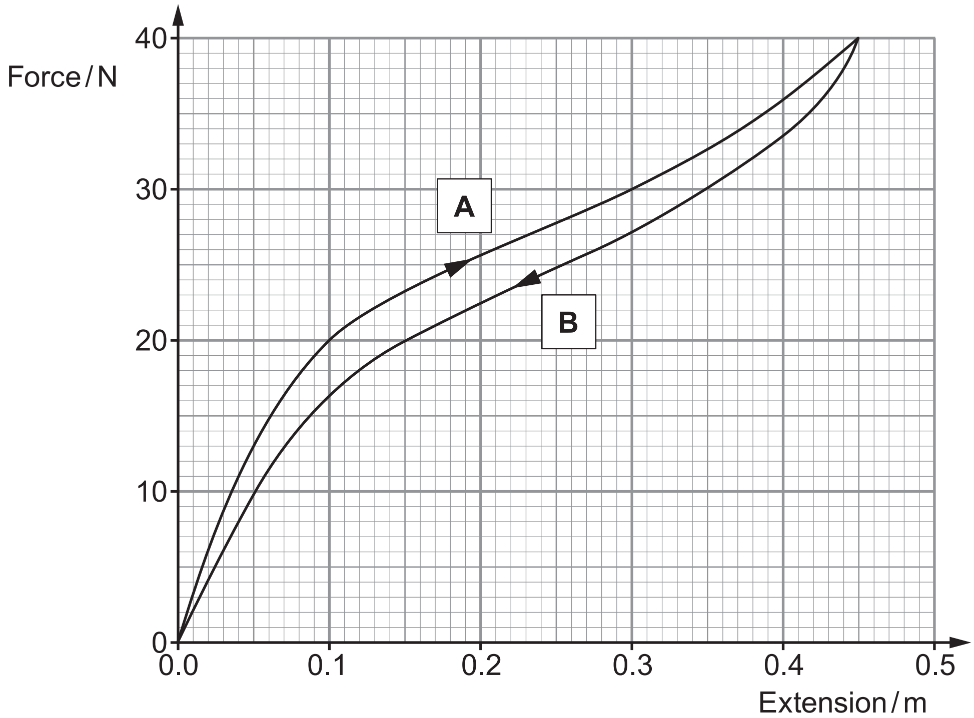
(b) Experiments are carried out on the rubber used in the bungee cord. The graph of load against extension is shown for a short piece of the cord. Curve **A** shows loading, and curve **B** shows unloading of the cord.



(i) State which feature of the graph confirms that the rubber cord is elastic. [1]

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(ii) State what happens to the molecules when the elastic band is stretched. [1]

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(iii) Showing your method, use curve **A** to estimate the work done in producing an extension of 0.3 m. [2]

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Examiner only

(iv) When the specimen is gradually unloaded, it is noted that the curve for unloading, **B**, is different from the curve for loading, **A**. Name this phenomenon **and** account for it in terms of energy. [2]

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